

Albert “AJ” Snow

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Chicago, IL

Senior Software Engineer

Senior Software Engineer with 7+ years of experience designing, developing, and supporting scalable backend and full-stack software systems. Strong background in software architecture, API and microservice development, database design, ETL pipelines, cloud infrastructure, DevOps, and automated testing across complex, high-volume environments.

Experienced with AWS, Python, JavaScript, Java, Clojure, SQL, Docker, Kubernetes, and CI/CD, with a strong focus on building reliable, secure, and maintainable systems.

Expertise: Software Development, Distributed Systems, Cloud Architecture, Technical Leadership, System Scalability, High Availability, Performance Optimization, Security Engineering, Software Reliability, Production Engineering, Technical Design, Requirements Analysis, Cross-Functional Collaboration, Code Quality, Test Strategy, Continuous Improvement, Incident Management, Root Cause Analysis, Agile Development, Software Development Lifecycle

Recent Professional Experience

Senior Software Engineer II

June 2025 – Present

Capital One

- Directed security compliance across 30+ services by tracking control status, coordinating remediation, and preparing audit evidence, achieving successful audits with 0 critical findings
- Spearheaded a cross-team service migration through completion while resolving 3 critical production incidents associated with the transition
- Key Accomplishments:**
 - Partnered with clients to scope requirements and collaborated with frontend engineers to build a serverless AWS self-service tool from scratch, eliminating 16 hours of manual work per month
 - Migrated a core service from a legacy HashiCorp Vault to AWS Secrets Manager, eliminating vulnerability surface area and reducing our monthly vulnerabilities count by 60
 - Reduced on-call burden by 8–10 hours per month by eliminating false positives, consolidating PagerDuty structures, and resolving root causes for persistent issues
 - Modernized domain failover from a multi-hour manual process to a minutes-long workflow and led the fastest region failback to date

Senior Software Engineer

February 2024 – June 2025

Center for Translational Data Science (CTDS) at University of Chicago

- Strengthened reliability across 6 services by developing 250+ tests and resolving logging anomalies to improve behavioral consistency
- Maintained FedRAMP Moderate compliance across new code contributions, including the search-saving API, while aligning implementation with required security controls
- Supported reliability for 30+ active services through on-call ownership, including code reviews, production debugging, and pipeline remediation
- Key Accomplishments:**
 - Architected and owned a search-saving API from design through production release, achieving zero reported bugs post-launch
 - Created an AI accuracy-scoring system which incorporated valid string rephrasings in scoring as part of a research initiative intended to reduce data ingestion turnaround from weeks to days

Back-End Engineer III

April 2022 – June 2023

Peerspace

- Translated stakeholder requirements into 2 technical design documents, establishing clear technical scope and implementation strategies for new capabilities
- Optimized database architecture by adding critical fields to the core schema while removing 20+ redundant fields across MongoDB schemas
- Diagnosed software faults by systematically analyzing and filtering approximately 100K–1M system logs to identify production issues and underlying causes
- Key Accomplishments:**
 - Launched an external calendar synchronization capability that achieved 30% adoption within its first 2 weeks and earned strong client feedback

- ▶ Established TDD practices by introducing 130+ tests and helping define new company procedures for integrating automated testing into service development

Software Engineer II

June 2019 – March 2022

Paycom

- Collaborated with a 17-person engineering team to deliver software within a 500K+ line-of-code domain while navigating complex system dependencies
 - Safeguarded sensitive data and software delivery by adhering to ISO 9001:2015, SOX, and PII requirements throughout development and data storage
 - Expanded backend capabilities by developing 180+ API endpoints, user-facing scripts, and ETL workflows supporting core business functionality
 - **Key Accomplishments:**
 - ▶ Drove design of a core dynamic job-field customization capability adopted by 80% of 10K+ clients
 - ▶ Streamlined development costs and maintainability by abstracting repetitive code patterns, reducing one implementation from approximately 1,000 lines to 200
 - ▶ Accelerated client-facing performance by optimizing application logic, aggregating MySQL queries, and tuning ORMs to reduce wait times by up to 100x
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Education, Certifications, and Technologies

Education: Bachelor of Science, Computer Science, Kansas State University

Certifications: AWS Senior Solutions Architect (Associate)

Languages: Python, JavaScript, Java, Clojure, PHP, Rust, C#, C++

Tools: AWS, Node/Express, MySQL, Docker, Splunk, GCP, New Relic, PostgreSQL, Kubernetes, Spring Boot

Skills: Software Architecture, Database Design, API/REST, CI/CD, MVC, Microservices, System Design, ETL, OOP, DevOps, Backend, Full-Stack, Leadership, Technical Problem Solving